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welche uni angeben, welche reihenfolge..?

Automatic Optimization of Level Parameters for RSHash

(RSHash is a minimizer based fast, compressed hash table for storing DNA k-mers. It was recently introduced by the Reinert Group.)

Bachelor's Thesis

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Berlin, _____ _____

Date Signature

Abstract

in English

Zusammenfassung

auf Deutsch

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1 Introduction

1.1 RSHash Explanation

Wie detailiert soll ich RSHash beschreiben? Reicht eine grobe Beschreibung und ein Verweis auf [SMR26]?

RSHash is a minimizer based fast, compressed hash table for storing DNA k-mers that answers lookup and can be extended for supporting locate. It was recently introduced by the Reinert Group.
The data structure works with one, two or three levels and a hash table. For each level i a pair of level parameters has to be specified: the minimizer length m_i and a threshold t_i.
Each level i works in the following way: The minimizer of length m_i is computed for each super-k-mer in the input whilst also counting the frequency (occurrences) of all appearing minimizers.
All super-k-mers with minimizers whose frequency does not succeed t_i are marked in a bit vector R, their offsets in the Text are stored in a vector O and a bit vector S marks the boundaries of the offsets for each minimizer.
Those super-k-mers with minimizers of a frequency larger than t_i make up the input for the next higher level $i + 1$ which works exactly like the previous apart from having its own level parameters.
Only super-k-mers with frequent k-mers on the highest level specified end up being stored in the hash table.[SMR26].

1.2 Problem Definition

Choosing the minimizer length **m** is an inherent problem of minimizer based k-mer indexing methods. A minimizer is expected to be valid for an average of $(k - m + 2)/2$ k-mers. Using a small **m** thus leads to a relatively small number of super-k-mers meaning a better compression. Picking a small **m** however also means that minimizers will be more frequent which leads to high query times because each minimizer offset has to be evaluated for an actual match with the queried k-mer. This means a constant trade off between space consumption and query times of minimizer based k-mer indexing methods. Existing methods use a number of approaches to deal with this problem (beispiele und quellen angeben)

The leveled approach used by RSHash exploits the skewed distribution of minimizers. It allows to achieve high compression by using relatively short minimizers on the lower

levels whilst also mitigating the burden of very frequent minimizers by using thresholds.

So far the level parameters have to be specified when building the data structure which requires manual testing and extensive understanding of the RSHash algorithm.

With this thesis we attempt expand the understanding of how exactly the level parameters influence the time and space consumption of RSHash and make the algorithm more accessible to use by automating the optimization of level parameters for a given genome.

Problems:

- work towards a better understanding of minimizer distribution across several levels
- analyze how level parameters influence time and space consumption of RSHash
- automize finding optimal level parameters

2 Minimizer Distribution

The skewness of minimizer distributions has been observed by several previous works [Chi+14; Jai+20; Pib22].
We will focus on the percentages $\mathbf{p_M}$ of minimizers with frequency at most f by minimizer length m and $\mathbf{p_K}$ of k-mers having those minimizers as described in [SMR26].

und > 0?

soll ich hier tatsächlich auch p_M einführen, obwohl ich p_M bisher nirgends verwende? ich denke schon, da p_M hilfreich ist, um p_K zu verstehen

We aim to analyze how the minimizer distribution behaves across the several levels used by RSHash in order to gain a better understanding of how to chose the level parameters.

2.1 Minimizer Distribution for Several Levels

We assume the distribution of minimizers on higher levels to be dependent on the choice of minimizer lengths and thresholds on the levels below, as those influence which k-mers appear on higher levels.
In order to understand this relationship, we will measure these distributions for a selection of sequences and analyze the results. It should be noted that this alone is not sufficient to argue about the general case.

1.

3 MINLP Modeling

We introduce a Mixed Integer Non Linear Programming (MINLP) aiming to optimize the level parameters. First we model the runtime and space needed by RSHash on One Level in dependence of the level parameters and minimizers distribution. We then extend this to the general case of Several Levels.
For a given Sequence we can then compute the minimizer distribution as outlined in Minimizer Distribution and then use a MINLP solver to find those level parameters leading to the optimization goal.

3.1 Notation

Let q be a k-mer. We refer to its minimizer $m(q)$ and frequency $f(m(q))$.
We denote the time needed to perform an operation *op* on a data structure *struct* with $T_{op_{struct}}$ - e.g. the time for a rank query on a vector V would be T_{rank_V}
We use $\mathcal{M}$ and $\mathcal{T}$ to denote the sets of minimizer lengths and thresholds considered when searching for optimal parameters.
We denote the optimal minimizer length $\mathbf{m}$ and threshold $\mathbf{t}$.
Closely following the notation introduced by [SMR26] $\mathbf{p_K}$ denotes the percentage of appearing k-mers q with infrequent minimizers of length $\mathbf{m}$ ($f(m(q)) \leq \mathbf{t}$). $p_K(m,t)$ on the other hand is the empirically measured percentage on a fixed sequence and a given pair $(m,t) \in \mathcal{M} \times \mathcal{T}$.
Similarly $\mathbf{f_{avg}}$ describes the average frequency of all infrequent minimizers, whilst $f_{avg}(m,t)$ is the empirically measured average frequency for a specific sequence and a given pair $(m,t) \in \mathcal{M} \times \mathcal{T}$. $f_{avg}(m,t)$ can be derived from $p_K(m,t)$ using dynamic programming:

model these times as part of the optimization

$$
\begin{aligned}
f_{avg}(m,t) &= \sum_{f=1}^{t} \frac{(p_K(m,f) - p_K(m,f-1))}{p_K(m,t)} \cdot f \\
&= \frac{1}{p_K(m,t)} \cdot \sum_{f=1}^{t} (p_K(m,f) - p_K(m,f-1)) \cdot f
\end{aligned}
$$

tabelle $f_avg(m,t)$ mittels dynamischer programmierung bauen

3.2 One Level

Runtime

Times for Operations on Datastructure

Vector Access Accessing an element of a Vector can be done in constant time, we use the following values based on measurements:

$$T_{access_R} = ...$$
$$T_{access_O} = ...$$

Hash Table Access Accessing an element of a Hash Table can be done in constant time, we use the following values based on measurements:

$$T_{access_H} = ...$$

Bit Vector Rank Query A bitvector of length U with n $1s$ can be encoded with $n\lceil logU/n \rceil + 3n$ many bits supporting rank in $O(log(U/n)$ time (Okanohara and Sadakane, 2007).

does that help here? Doesn't the model require actual time instead of O notation - maybe measurements would be more suited

Bit Vector Select Query ???

Comparison with Text ???

Lookup

A k-mer q is looked up by first computing its minimizer $m(q)$. If the bit $R[m(q)]$ is not set, q is not in the data structure.
For the $\mathbf{p_K}$ of the k-mers for which the bit is set, the minimizer's rank $r = rank_R(m(q))$ is computed. Then $select_S(r)$ and $select_S(r+1) - 1$ tell the indices in Offsets O where the minimizer's first and last offset in the text T are stored.
Finally the presence of q in T is checked at every offset $o \in O[select_S(r) : select_S(r + 1) - 1]$.
This offset interval contains an average of $\mathbf{f_{avg}}$ offsets. In the lookup case, the comparison is terminated as soon as q is found for the first time, which is on average the case after $(\mathbf{f_{avg}} + 1)/2$ comparisons (+1 is necessary because the first index always has to be checked).
For $1 - \mathbf{p_K}$ $m(q)$ does not appear at all or has a $f(m(q)) > t$. In this case the lookup is completed through an access of the hash table.[SMR26]

The lookup time $T_{lookup_{RSHash}}$ for one level can thus be modeled as:

$$T_{lookup_{RSHash}} = \begin{aligned} & T_{access_R} + \mathbf{p_K} \cdot (T_{rank_R} + 2T_{select_S} + (\mathbf{f_{avg}} + 1)/2 \cdot (T_{access_O} + T_{compare_T})) \\ & + (1 - \mathbf{p_K}) \cdot T_{access_H} \end{aligned}$$

In order to determine m and t we introduce binary selector variables μ_m, τ_t for each possible $m \in \mathcal{M}$ and $t \in \mathcal{T}$.

sollte es $max(\frac{f_{avg}}{2}, 1$ sein?

μ_m should be 1 exactly when the optimal minimizer length is m.
τ_t should be 1 exactly when the optimal threshold is t.
The following constraints force the selectors for m and t to be binary and mutually exclusive:

$$\begin{aligned} \mu_m &\in \{0,1\} \mid \forall m \in \mathcal{M} \\ \tau_t &\in \{0,1\} \mid \forall t \in \mathcal{T} \\ \sum_{m \in \mathcal{M}} \mu_m &= 1 \\ \sum_{t \in \mathcal{T}} \tau_t &= 1 \end{aligned}$$

With $p_K(m,t)$ being the empirically determined percentage of k-mers q with $1 \leq f(m(q)) \leq t$ (for a fixed $(m,t) \in \mathcal{M} \times \mathcal{T}$) we can then model $\mathbf{p_K}$ with:

$$\mathbf{p_K} = \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} p_K(m,t) \cdot \tau_t \right)$$

which takes the value of the only $p_K(m,t)$ for which $\mu_m == 1$ and $\tau_t == 1$.

Analogously with $f_{avg}(m,t)$ being the empirically determined average frequency of k-mers q with $1 \leq f(m(q)) \leq t$ (for a fixed $(m,t) \in \mathcal{M} \times \mathcal{T}$) we can then model $\mathbf{f_{avg}}$ with:

$$\mathbf{f_{avg}} = \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} f_{avg}(m,t) \cdot \tau_t \right)$$

The above results in the following MINLP:

$$\begin{aligned} &\text{MINIMIZE} \\ T_{lookup_{RSHash}} &= T_{access_R} + \mathbf{p_K} \cdot (T_{rank_R} + 2T_{select_S} + (\mathbf{f_{avg}} + 1)/2 \cdot (T_{access_O} + T_{compare_T})) \\ &\quad + (1 - \mathbf{p_K}) \cdot T_{access_H} \\ &\text{SUBJECT TO} \\ T_{access_R} &= \\ T_{rank_R} &= \\ T_{select_S} &= \\ T_{access_O} &= \\ T_{compare_T} &= \\ T_{access_H} &= \\ \mathbf{p_K} &= \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} p_K(m,t) \cdot \tau_t \right) \\ \mathbf{f_{avg}} &= \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} f_{avg}(m,t) \cdot \tau_t \right) \\ \sum_{m \in M} \mu_m &= 1 \\ \sum_{t \in T} \tau_t &= 1 \\ \mu_m &\in \{0,1\} \mid \forall m \in M \\ \tau_t &\in \{0,1\} \mid \forall t \in T \end{aligned}$$

Solving this MINLP will result in a Solution where the μ_m, τ_t corresponding to the smallest $m \in \mathcal{M}$, $t \in \mathcal{T}$ will be set to one.
This is expected, as this leads to the smallest possible $\mathbf{p_K}$ which minimizes the k-mers having to be looked up in the text and maximizes the lookups in the (faster) hash table. This however results in maximal space consumption, as each entry in the hash table is significantly larger than an entry in O as described in Problem Definition.

In order to account for that, we have to add constraints modeling the space consumption.

Space Consumption

The space consumption of RSHash can be decomposed into the space needed for the data structures components: the vectors R, O and S, the text T as well as the hash table H.

Bit Vectors R and S

R and S can likely be disregarded when modeling the space consumption.
R is a Bit Vector of length $4^{\mathbf{m}}$ that contains n 1s marking the existence of each encountered minimizer. Using Elias-Fano-Encoding, its size can be compressed to

explain why

possibly model n?

$$n\lceil log(4^{\mathbf{m}}/n)\rceil + 3n$$

Offsetvector O

O contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t$ in T.
Every minimizer covers an average of $(k - \mathbf{m} + 2)/2$ consecutive k-mers. We can thus model the bits per k-mer in O as:

$$bpk_O = \frac{\mathbf{p_K}}{(k - \mathbf{m} + 2)/2} \cdot log|T|$$

Hash table H

For $1 - \mathbf{p_K}$ of the k-mers q in T H contains an entry $(key : q, value : offset(q))$
Each entry has size $size(key) + size(value) = 64 + log|T|$. We introduce the load factor α to the model, in order to account for empty buckets in the hash table.
The bits per k-mer required for the hash table can thus be modeled by

$$bpk_H = (1 - \mathbf{p_K}) \cdot \frac{(64 + log|T|)}{\alpha}$$

3.3 Several Levels

As explored in Minimizer Distribution, we assume the minimizer distribution on higher levels to be practically identical to the distribution on the first level.

falsch!!
Anpassungen überlegen

Runtime

RSHash uses several levels, to make use of the skewniss of minimizer distribution.
Each level l has a pair of parameters $\mathbf{m}^l, \mathbf{t}^l$ which result in a probability $\mathbf{p}_\mathbf{K}^l$ of a k-mer belonging to level l.
For now, we still assume, that the time for a lookup in the vectors R, S as well as the hash table H are (practically) independent of the parameters $\mathbf{m}^l, \mathbf{t}^l$.

reason

When using L levels, the probability of a k-mer belonging to a higher level is dependent on the probability of k-mers belonging to the levels below.
On each level $1 \leq l \leq L$, for $\mathbf{p}_\mathbf{K}^l$ of the k-mers, the lookup will be completed by finding a value in the offset vector on that level whilst the remaining $(1 - \mathbf{p}_\mathbf{K}^l)$ queries result in a lookup on the next level.
We can model the lookup time for all levels $1 \leq l \leq L$ levels recursively as:

$$T^l_{lookup_{RSHash}} = \begin{cases} T_{access_{R^l}} + \mathbf{p}_\mathbf{K}^l \cdot (T_{rank_{R^l}} + 2T_{select_{S^l}} + \dfrac{\mathbf{f}^l_\mathbf{avg}}{2} \cdot (T_{access_{O^l}} + T_{compare_T})) & | \ 1 \leq l \leq L \\ \quad + (1 - \mathbf{p}_\mathbf{K}^l) \cdot T^{l+1}_{lookup_{RSHash}} & \\ T_{access_H} & | \ l = L + 1 \end{cases}$$

As an example for $L = 2$ levels, this results in:

$$\begin{aligned} T_{lookup_{RSHash}} = {} & T_{access_{R^1}} + \mathbf{p}_\mathbf{K}^1 \cdot (T_{rank_{R^1}} + 2T_{select_{S^1}} + \frac{\mathbf{f}_\mathbf{avg}}{2} \cdot (T_{access_{O^1}} + T_{compare_T})) \\ & + (1 - \mathbf{p}_\mathbf{K}^1) \cdot \Big(T_{access_{R^2}} + \mathbf{p}_\mathbf{K}^2 \cdot (T_{rank_{R^2}} + 2T_{select_{S^2}} + \frac{\mathbf{f}_\mathbf{avg}}{2} \cdot (T_{access_{O^2}} + T_{compare_T}) \\ & + (1 - \mathbf{p}_\mathbf{K}^2) \cdot T_{access_H} \Big) \end{aligned}$$

Because $\mathbf{m}^l$ and $\mathbf{t}^l$ can be different on every level l, a different set of binary selector variables μ^l_m and τ^l is needed.
The constraints for each level are the same as for one level:

$$\begin{aligned} \mu^l_m &\in \{0,1\} && | \ \forall (l,m) \in (\{1,...,L\} \times \mathcal{M}) \\ \tau^l_t &\in \{0,1\} && | \ \forall (l,t) \in (\{1,...,L\} \times \mathbf{t}) \\ \sum_{m \in \mathcal{M}} \mu^l_m &= 1 && | \ \forall l \\ \sum_{t \in \mathcal{T}} \tau^l_t &= 1 && | \ \forall l \end{aligned}$$

For every level l $\mathbf{p}_\mathbf{K}^l, \mathbf{f}_\mathbf{avg}^l \mathbf{m}^l$ and $\mathbf{t}^l$ can also be modeled analogously to having only one level:

$$\mathbf{p}_\mathbf{K}^l = \sum_{m \in \mathcal{M}} \left(\mu_m^l \sum_{t \in \mathcal{T}} p_K(m,t) \cdot \tau_t^l \right) \quad | \forall l \in \{1, ..., L\}$$

$$\mathbf{f}_\mathbf{avg}^l = \sum_{m \in \mathcal{M}} \left(\mu_m^l \sum_{t \in \mathcal{T}} f_{avg}(m,t) \cdot \tau_t^l \right) \quad | \forall l \in \{1, ..., L\}$$

$$\mathbf{m}^l = \sum_{m \in \mathcal{M}} \mu_m^l \cdot m \quad | \forall l \in \{1, ..., L\}$$

$$\mathbf{t}^l = \sum_{t \in \mathcal{T}} \tau_t^l \cdot t \quad | \forall l \in \{1, ..., L\}$$

Space Consumption

Offsetvector O

For several levels, the offset vector for level l only contains an entry for a k-mer q, if the k-mers minimizer $m(q)$ has a frequency $f(m(q)) > t^i$ for all levels $1 \leq i < l$ and $f(m(q)) \leq t^l$.
The bits per k-mer for the Offset vector on level l can thus be modeled as:

$$bpk_{O^l} = \prod_{i=1}^{l-1} (1 - \mathbf{p}_\mathbf{K}^i) \cdot \mathbf{p}_\mathbf{K}^l \cdot \frac{2}{k - \mathbf{m}^l + 2} \cdot log|T|$$

Hash table H

The hash table only contains an entry k-mers q, for which $f(m(q)) > t^l$ for all levels $1 \leq l \leq L$.
The bits per k-mer for the hash table can thus be modeled as

$$bpk_H = \prod_{l=1}^{L} (1 - \mathbf{p}_\mathbf{K}^l) \cdot (64 + log|T|)$$

Limitations

This approach of solving the constructed model has the advantage () of finding optimal results .
However for large genomes, finding the empirical values $p_K(m,t)$ becomes increasingly time consuming and needs a lot of RAM .

paragraph/section about the complexity of finding the solution

For these reasons, an approach yielding a solution that is a certain approximation of the optimum but takes less time to find it could be of interest.

this only holds under the assumption, that the minimizer distribution is equal on each level - independent of the choice of m and t. This is not trivial and should be further explored!!

should I introduce an index for the minimizer function as it differs for every level? e.g. $m^l(q)$

over what

does it?

how

An easy adaption to speed up the search for empirical values would be to only sample part of the genome. By sampling sequences of length at a large number of random locations, the probability of choosing a bad representation becomes highly unlikely.
By sampling only part of the genome, we can speed up the finding of empirical values by a factor of / or even "get it down to a constant" (that is for a fixed number of sample locations - regardless of the genomes size. this however results in a growing error for larger genomes).

what length

how lorge

factor

4 Subsampling

Motivation

The time for processing a genome in order to find the minimizer distribution $\mathbf{p_K}(m, t)$ for possible minimizer lengths $m \in \mathcal{M}$ and thresholds $t \in \mathcal{T}$ is directly proportional to the size of the genome.
For large genomes, this processing can become the dominating factor when searching for optimal parameters using the MINLP Modeling approach. In order to speed up the search for the empirical values, we introduce a subsampling method.

Subsampling Method

Instead of calculating the distribution over the full genome, we sample n sequences of length $l \geq k$ at random positions in the genome.

how to handle overlapping samples

Time Complexity of Finding the Minimizer Distributions

genaue zeitliche abhängigkeit 1. verstehen und 2. aufschreiben

Time Complexity using Subsampling

By applying the proposed subsampling strategy, the original genome T of length $|T|$ is reduced to the subsampled text T^S of length $|T^S| = n \cdot l$. This means a reduction in size by a factor of $\frac{|T|}{|T^S|}$.
Depending on the choice of n and l this results in a speedup of the processing time by a factor of or even a constant processing time when both parameters are chosen as a constand regardless of $|T|$ (this however should result in a growing error for larger genomes).

assuming no overlaps

factor

time

Error Analysis

Choosing l

The length l has to be at least k.

wenn $l > k$ ist, überlappen sich mehrere kmere an den gesampleten stellen. ich verstehe noch nicht, warum genau dass von vorteil ist.
Weil roling hash - l ca 1000, dann quasi linear

Choosing n

Choosing a constant n would result in a processing time that would no longer be dependent of the genomes length $|T|$, as the sampled text would have constant size $n \cdot l$. For large genomes, the percentage $\frac{|T^S|}{|T|}$ of the genome represented in the subsample can become arbitrarily small, resulting in growing error.
If instead n is chosen relative to $|T|$, $\frac{|T^S|}{|T|}$ will have a constant value independent of the genomes size. Even though this means, that the time needed, in order to find the minimizer distributions would still be proportional to the genomes size, this should be the preferred option as it results in a bounded error of the approximation.
For a large enough n, the probability of choosing a bad representation becomes highly unlikely.

assuming no overlaps

zu vage, begründen

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